
GPU-Parallelizing Arbitrary Python Code By Running 1 Million Python Interpreters on a GPU

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Abstract

We introduce *gpusnek*, a fully functional Python interpreter ported to CUDA, enabling execution of arbitrary Python code directly on the GPU by running one whole interpreter on every CUDA core/thread. This is a tremendously bad idea, but for the duration of this paper we pretend that it is not, focusing instead on the surprisingly good job it does of combining the massively parallel compute capabilities of the GPU with the expressive power and ease-of-use of Python. Our implementation of *gpusnek* exposes the full CUDA memory hierarchy to Python in addition to key GPU-specific thread synchronization primitives, which allows serious GPU-native parallel algorithms to be written and run in pure Python. It also makes available many Python-native features that are traditionally difficult to achieve directly from a GPU thread, like reading from / writing to a filesystem. There are many possible applications of the *gpusnek* library (yes, we have made it available as a library). But in this work we choose to use it to build an interactive Python REPL backed by 1 Million interpreters running on a single GPU, unlocking instant and effortless million-way parallelism for your interactive Python terminal sessions.

<https://github.com/jndean/gpusnek>

If you just want to see the fun examples, skip to Section 2.

1. The Folly of Others

There are many serious projects that try to sell you “Python but on the GPU”, and these tend to be doing one (or, often, both) of the following two things. Firstly, there are those projects that run Python code on the host CPU, and that Python code is used to plug together and schedule pre-written GPU kernels on the device (e.g., cuPy, PyTorch). In this case, the code running on the GPU is most definitely not Python; it is largely hand-optimized machine code provided by NVIDIA and other vendors to perform common vector-

```
gpusnek REPL; gpu with 1,048,576 snek
>>> print('I am thread', threadIdx % 3)
----- 349526x -----
I am thread 0
----- 349525x -----
I am thread 1
----- 349525x -----
I am thread 2
>>>
```

Figure 1. Running a print statement in an interactive Python REPL backed by 2^{20} Python interpreters. Output is deduplicated and printed with blue multiplicity banners, showing three unique responses each printed $\sim 350K$ times.

ized linear algebra operations, and woe betide any workload that cannot be expressed in terms of the available operations! The second capability offered by existing projects is to use Python to write the GPU kernel itself, either in the traditional per-thread execution model (e.g., Numba, Mojo), or by offering vector constructs within Python’s syntax that express block-level logic (Triton, cuTile, Palas). In this case, too, it is most definitely not Python that is actually running on the GPU. Code written with these tools is transliterated, traced, JIT-compiled, and otherwise transmogrified into real device-specific machine code before execution. These magical code-gen pipelines offer you the regular kernel programming experience hidden behind the familiar-looking syntax of Python, and generally restrict the programmer to a very limited subset of Python’s features and data types. Thus we find them to be lacking in the real power (and pain) that a true dynamically-typed and interpreted language should achieve when executing live on the GPU.

In contrast, *gpusnek* has the gall to put the whole Python interpreter directly on the GPU. Not just some for loops and some numerical data types, but the full language spec. Not just the bytecode-interpreting virtual machine, but also the lexer, the parser and the compiler, so your threads can even write source code as strings and send them directly to other threads to be parsed and executed. Not just a set of isolated single-threaded execution sandboxes, but a runtime

with access to a real file system, the GPU’s explicitly managed L1 cache, and inter-interpreter thread communication primitives that might let one get some real work done.

2. Examples

When one can execute arbitrary Python code on the GPU, what should one do with it?

2.1. Hello World

Figure 2 shows a GPU kernel that accepts a buffer of space in which each thread initializes a separate *gpusnek* interpreter. Then, this interpreter executes a Python source code string embedded into the kernel binary, printing “Hello World!”.

```
__global__
void my_kernel(char* mem, int per_thread_mem) {
    char* my_mem = &mem[threadIdx.x * per_thread_mem];
    gpusnek_init(my_mem, per_thread_mem);

    gpusnek_do_str("print('Hello World!')\n")
}
```

Figure 2. A basic *gpusnek* kernel that prints a string directly to *stdout*.

Easy! Notice how nothing needs to be passed from the *gpusnek* initialization to subsequent *gpusnek* functions; once an interpreter has been set up by a thread, it continues to be accessible by the same thread forever after (assuming that the working memory is still alive), even across other kernel launches. For that reason, the rest of our examples omit the *gpusnek_init* call, assuming that a previous kernel launch has already set up the working memory ready for us to use. *gpusnek_do_str* utilizes the interpreter to execute a string of Python source code that is embedded into the binary. It is possible to instead embed and run pre-compiled MicroPython bytecode, but since we have also ported a GPU-native Python lexer and parser, we might as well use them to make the examples readable!

2.2. Co-operating Threads

Next we explore a few *gpusnek* features that make it possible to get real work done. Figure 3 shows a kernel for computing the mean of an array using a parallel reduction algorithm in $\log_2(N)$ time. It demonstrates the *gpusnek* API for making C++ data available inside the Python interpreter; the *gpusnek_new_int* method just injects a new Python integer into the global namespace, while *gpusnek_bind_memory* makes a region of memory available as a Python MemoryView object with configurable data types. Specifically in this case, we bind *data* to the data buffer in global HBM while *cache* points to a region of the L1 cache local to the thread block. This allows the Python code to perform a nice contiguous data load from HBM and then do the reduction

entirely in the cache. The kernel also uses the *synctreads()* ‘function’, which is one of our special additions to the MicroPython runtime (see Section 3.2.1). Rather than being a true function call, it gets mapped directly by the on-device compiler to a single bytecode instruction that invokes the *__synctreads()* CUDA intrinsic, acting as a thread barrier that syncs all CUDA threads within the same block.

```
__global__
void average_kernel(float *data) {
    int threadId = threadIdx.x;

    __shared__ float cache[NUM_THREADS];
    gpusnek_bind_memory("cache", cache, NUM_THREADS, 'f');
    gpusnek_bind_memory("data", data, NUM_THREADS, 'f');
    gpusnek_new_int("tid", threadId);
    gpusnek_new_int("N", NUM_THREADS);
    gpusnek_do_str(R"

from math import log

cache[tid] = data[tid]
synctreads()

steps = int(log(N, 2))
for i in range(steps):
    offset = N >> (i + 1)
    if tid < offset:
        cache[tid] += cache[tid + offset]
    synctreads()

if tid == 0:
    data[0] = cache[0] / len(data)
)");
}
```

Figure 3. A kernel that takes the mean of an array using a parallel reduction sum, writing the answer into element 0. The kernel should be launched with one thread per element of data.

2.3. An In-Memory Virtual Filesystem

Example 2.2 showed the use of Python to implement a very GPU-shaped workload. Next, we divert to something very Python-native and entirely not GPU-shaped: reading and writing strings to files.

Figure 4 shows a simple example kernel that starts by binding a globally shared memory buffer to each interpreter. Next, it creates a block device which mimics storage hardware (such as flash storage) that accepts data read/write requests in blocks. In reality, this virtual block device is just a Python object that reads and writes the data to the shared memory buffer. Thread 0 then formats this ‘device’ as a *littlefs2* filesystem, and both threads mount it. From here on, they have access to a shared filesystem that lives entirely in the GPU VRAM, able to do all the usual things one expects to do with a filesystem. One thread could write Python code to a file and another could import and execute it as a module using a standard Python *import* statement! But in this example, thread 1 simply reads the contents of a file written by thread 0.

```

__global__
void fs_kernel(char *fs_buf, int fs_size) {
    gpusnek_new_int("threadId", threadIdx.x);
    gpusnek_bind_memory("fs_buf", fs_buf, fs_size, 'B');
    gpusnek_do_str(R"(
import vfs

# Definition of VRAMBlockDevice omitted for brevity
bdev = VRAMBlockDevice(
    block_size=128,
    buffer=fs_buf,
)

if threadId == 0:
    vfs.VfsLfs2.mkfs(bdev)
    vfs.mount(bdev, '/vfs')
    message = 'THE FILESYSTEM LIVES ON THE GPU'
    with open('/vfs/myfile.txt', 'w') as f:
        f.write(message)

syncthreads() # Is a barrier

if threadId == 1:
    vfs.mount(bdev, '/vfs')
    with open('/vfs/myfile.txt', 'r') as f:
        print(f.read()) # Prints the message
)");
}

```

Figure 4. Thread 0 initializes a shared VRAM buffer as a disk image for a virtual filesystem. Both threads mount this filesystem, and use it to pass a message between them.

2.4. One Million Python Interpreters (the Mega-REPL)

Next we demonstrate the scale of the parallelism available on a GPU. MicroPython provides an excellent pre-made REPL (Read-Eval-Print Loop) that lets a user write and execute Python code interactively in a terminal. Once the interpreter is initialized, the provided `pyexec_event_repl_process_char` function can be called to ingest one character at a time and, after each call, potentially print output from an execution (execution is likely only triggered when a return character is ingested). Having ported this MicroPython routine to CUDA as a part of *gpusnek*, we can have the host capture one key press at a time and send them via individual GPU kernel launches to be ingested by one (or many) GPU-side Python REPLs. The output from all interpreters is sent back to the host and de-duplicated, then printed to the terminal with multiplicity (see the turquoise banners in Figure 5). In this way, we create an interactive REPL where code is executed by as many interpreters as we can fit on the GPU, and all of the outputs are presented back to the user. By allocating just 12KB to each heap, 5KB to each stack, and 1KB to each *stdout* buffer, we can comfortably fit 2^{20} interpreters on an NVIDIA RTX 4070 Ti SUPER with 16GB of onboard VRAM, and thus unlock instant interactive million-way parallelism in Python on a consumer desktop platform. Figures 5 and 6 attempt to convey the experience of using it.

```

Starting gpusnek REPL with 1,048,576 snek...
>>> numThreads
1048576
>>> threadIdx % 3
----- 349526x -----
0
----- 349525x -----
1
----- 349525x -----
2
>>>

```

(a) Starting up an interactive terminal session with the Mega-REPL and printing some runtime-provided values. Outputs are deduplicated and printed with turquoise multiplicity banners (unless they are all identical, in which case the banner is omitted).

```

>>> if int(threadIdx ** 0.125) ** 8 == threadIdx:
...     print('8th power:', threadIdx)
...
----- 1x -----
8th power: 0
----- 1x -----
8th power: 1
----- 1x -----
8th power: 256
----- 1x -----
8th power: 390625
----- 1x -----
8th power: 65536
----- 1x -----
8th power: 6561
----- 1048570x -----
>>>

```

(b) Using the effortless parallelism of the Mega-REPL to parallelize the search for 8th powers. Each thread only checks one value (its own thread index). 1, 048, 570 threads find and print nothing.

```

>>> with open('shakespeare.txt', 'r') as f:
...     f.seek(threadIdx * 6)
...     context = f.read(40)
...
>>> 'million' in context[:6 + len('million') - 1]
----- 1048556x -----
False
----- 20x -----
True
>>> if 'Snake' in context[:6 + len('Snake') - 1]:
...     print(context)
...
----- 1x -----
Snakes, in my heart-blood warm'd, that
----- 1048575x -----
>>>

```

(c) The interpreters have a shared a filesystem loaded into VRAM, and we have preloaded the complete works of William Shakespeare as a (~ 5.2 MB) text file. Each thread reads just a small chunk of the file (the full contents wouldn't fit in any one thread's heap!). Interactive parallelized searches over the full text are easily achieved by querying just the local chunk: we find 20 (case-sensitive) instances of the word 'million', and print the context around the sole instance of 'Snake'.

Figure 5. Example use cases for a REPL with 2^{20} threads.

```

>>> img_size = int(numThreads ** 0.5)
>>> X = threadIdx % img_size
>>> Y = threadIdx // img_size
>>> x0 = (X / img_size) * 0.13 - 0.62
>>> y0 = (Y / img_size) * 0.13 - 0.65
>>> x, y, i, max_iter = 0, 0, 0, 30
>>> while ((x*x + y*y) <= 4) & (i < max_iter):
...     x, y = x*x - y*y + x0, 2*x*y + y0
...     i += 1
>>> colour = i / max_iter
>>> outbuf[threadIdx * 3: threadIdx * 3 + 3] = [
...     256 * (1 - colour), # B
...     256 * (colour ** 2), # G
...     256 * colour,      # R
... ]
>>>

```

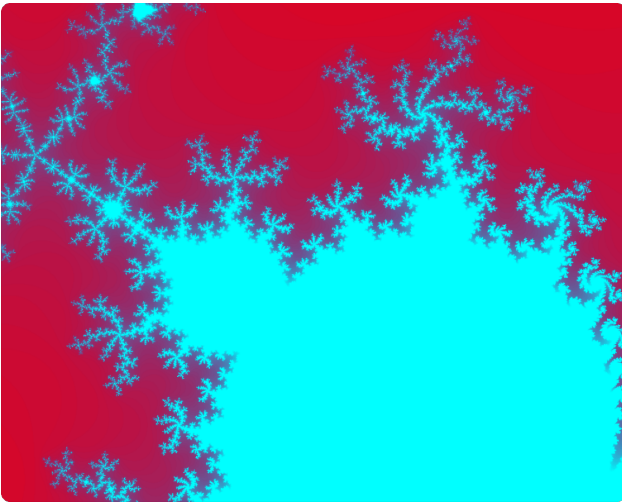


Figure 6. **(Top)** Each thread computes the colour of 1 pixel in the Mandelbrot Beetle fractal, and writes it into a buffer shared between all threads. **(Bottom)** Rendering that shared buffer as an RGB image texture. Using 2^{20} interpreters, each computing 1 pixel, quickly produces a 1024×1024 image (cropped here).

3. How It's Made

gpusnek is a port of MicroPython [1], which is a wonderful open source implementation of Python 3 designed to run on microcontrollers in embedded applications.

In some ways, running code on CUDA is a lot like running code on a lot of tiny microcontrollers, packed side-by-side onto a single chip. Thus, many of the design goals driving MicroPython make it ideally suited to our use case:

- **Small and lightweight** - MicroPython claims to need only 256K ROM and 16K RAM to operate, which is going to be useful if we want 1 Million independent copies living in the HBM of one GPU, running on relatively low-powered “cores”.
- **Configurable** - MicroPython defines a wealth of compile-time flags that allow us to disable problematic

subsystems or toggle between implementations that suit our hardware better. These days you can call *printf* on a GPU, but you still cannot call *malloc*!

- **Self-contained** - not relying on many external libraries improves compatibility with exotic / adversarially-chosen hardware and tool chains.

3.1. Porting Challenges

MicroPython has a well defined process for porting to a new hardware platform. The core systems are defined centrally, and they interact with the hardware by calling out to an API that must be implemented by the port-writer to, e.g., read a character from the keyboard or print to the terminal. However, in spite of MicroPython’s extensive configurability, our port to the GPU is sufficiently far from the beaten path that we must also make several invasive changes to the core Python implementation to get it working.

3.1.1. NVCC AND C++

To compile CUDA code to GPU machine code we must use NVIDIA’s C++ compiler *nvcc*. The first problem is that MicroPython is written in C99, not C++. The C language is *very nearly* just a subset of C++; however, when one shoves 60K lines of C code through a C++ compiler, one soon finds examples of the edge cases where that is not true. Unlike C, C++ does not like: *goto* statements jumping over variable declarations; struct initialization using field names; assigning void pointers to non-void pointers (which MicroPython does *a lot*); taking a reference to a temporary inline variable; and so on and so on. Fortunately, the compiler is very good at classifying and pointing out these problems, so the process of converting all of the core MicroPython systems to valid C++ code was quite laborious, but not (to our knowledge) perilous.

3.1.2. MUTABLE GLOBAL STATE

MicroPython is designed to be the only runtime on a single-threaded microcontroller, so it is able to store all of the interpreter state in static global mutable variables. This becomes a problem when running multiple copies of the interpreter on different threads on the same GPU - different threads executing different Python code will trample each other’s global state. The solution is clear: modify MicroPython to instead have a global *array* of interpreter states (dynamically initialized), and any accesses to these globals go through a macro that uses the current thread’s ID to index into the state array. This is why (as mentioned in Example 2.1) calls to the *gpusnek* API don’t need to be given a reference to the thread-local interpreter - each thread always knows how to look up its local interpreter using just its own thread ID.

3.1.3. SETJMP & LONGJMP

MicroPython makes use of the C functions *setjmp* and *longjmp*. The former allows you to checkpoint the current state of the CPU's registers into memory, while the latter allows you to reinstate a previously made checkpoint. Since the CPU registers tend to include the Program Counter (PC) and Stack Pointer (SP) (which track the current position in the program and the stack), calling *longjmp* acts like a *goto* statement that can jump out of functions, straight back to the location in the code where *setjmp* was called. MicroPython uses this functionality to implement exception handling by setting checkpoints at locations that can catch Python exceptions. When an exception is raised in a function that may be deep in the call stack, the runtime can *longjmp* to the latest checkpoint to unwind large numbers of stack frames, without having to define a way to return from all the intermediate frames in an error state. This is all very well, but unfortunately CUDA does not provide an implementation of *setjmp* and *longjmp*. To some degree, this might be because conceptually it does not map very well to GPU hardware. Below are some of the reasons we decided not to "just implement it ourselves in assembly".

- In modern GPUs, thread divergence is tracked explicitly by the program, which records bit masks at branch points in the code indicating which threads are still running in lock-step and are about to diverge. These masks are moved around as data, and may end up being stored on the stack if functions are called. When execution reaches an opportunity to reconverge threads, e.g., the end of an *if* statement block, the bit mask is read and used to determine which threads should be brought back in sync. If one of our Python threads participates in a thread divergence event but then raises an exception when its warp neighbours do not, that thread may jump over the thread reconvergence point in an intermediate call frame without participating, causing a deadlock when the other threads reach that point and wait for the masked sync.
- The above problem is an example of the somewhat arcane and also quite proprietary workings of the GPU architecture and its corresponding compiler. NVIDIA is not forthcoming with details about these systems, and it is only thanks to reverse-engineering efforts such as [3] that we can know enough to foresee the above issue. It is more than likely that other less foreseeable issues exist, either in software or in hardware, that would make resuming execution difficult to implement in "user-space", and a hack job implementation will just violate some opaque internal invariants and crash the GPU.
- While a CPU thread will have a handful of registers

holding the current logical state that need checkpointing by *setjmp*, a CUDA core can have hundreds of physical general purpose registers per thread holding state, and even more specialized registers. Moreover, NVIDIA's tools do not give us an easy way to enumerate and read from / write to all the registers currently being used by our thread, since developers are only really supposed to write a higher level assembly representation (PTX) that gets compiled through proprietary tools into the final executable. So, implementing *setjmp* would involve a non-trivial amount of manually assembling closed-source machine-code and find ways to link/inject it in to higher level kernels.

For the above reasons, we leave the implementation of *setjmp* and *longjmp* on the GPU as possible future work. Sections 3.1.5 and 3.1.4 touch more on the ways we work around their absence.

3.1.4. EXCEPTION HANDLING

Without *setjmp* and *longjmp*, MicroPython's exception handling mechanism cannot function. Currently in *gpusnek* we simply do not support exception handling: when an exception is raised, the exception message is printed directly to *stderr*, and the thread unceremoniously exits. As a workaround, we recommend writing Python code that does not raise any exceptions.

A partial solution would be possible if, as discussed above in Section 3.1.3, someone put the effort in to build an approximate *setjmp* implementation. There would still exist the problem that diverged threads shouldn't be allowed to jump in case they jump over reconvergence points, but the implementation could use CUDA intrinsic `__activemask()` to check for divergence of the warp at run time and succeed in cases which are probably safe.

We speculate that a full solution would be to undertake a major rewrite of MicroPython that makes exception handling a first-class member of the function-calling API, i.e., have it be explicit in every function. This sounds like a huge amount of work that would add run-time memory and cycle overhead, plus it could introduce copious boiler plate to the code. It's not even clear if it would work. As such, we do not consider this a direction for future work.

3.1.5. DYNAMIC MEMORY ALLOCATION AND GARBAGE COLLECTION

CUDA offers no dynamic memory allocation capability on the device, i.e., you cannot call *malloc* from a GPU thread. Instead, memory allocation is handled by the host CPU code, and those buffers are treated as static when passed in to GPU kernels. That's not going to work well for us if one of our GPU kernels contain Python interpreters that

expect to be able to dynamically create large data structures on-the-fly without ever talking to the host. Thankfully, MicroPython includes an implementation of a heap allocator, allowing us to pass a large static buffer in at interpreter initialization which will then be dynamically managed by the MicroPython runtime. As long as we give each thread its own pre-allocated heap space, this works well, with one small downside!

The MicroPython heap allocator uses a stack-sweeping garbage collection algorithm to reclaim unused memory. The garbage collector searches any locations that may contain root pointers to Python objects, and this includes the C call stack which may be holding references to local variables in a function call frame. In normal operation on a CPU, the exception handling mechanism (see Section 3.1.4) is also taking snapshots of the CPU's register state using C's `setjmp` function and saving them to the stack regularly, which means even object pointers not spilled from registers by the compiler will end up on the stack by garbage-collection time, ready to be swept. However, as we have already discussed, *gpusnek* is unable to take these register snapshots, and as such the callee-saved registers in each thread have a (rather slim) chance of being missed by the garbage collector. This is exacerbated by the tenacity with which the *nvcc* compiler attempts to avoid spilling registers to memory. We employ the following desperate mitigations to increase the chances that all live Python variables are referenced somewhere the garbage collector can find them:

- Separate compilation unit for the top-level garbage collection routine to reduce the likelihood the function call is inlined by *nvcc*.
- Enable MicroPython's builtin 'pystack' implementation which stores the Python call frames in a separate explicitly managed stack buffer rather than directly in the C call stack. The garbage collector can search this stack buffer.

To date, this seems to have been sufficient to prevent disaster.

3.2. Adding New Functionality

Having overcome the porting challenges described above in Section 3.1, *gpusnek* could have been considered a technically functional but minimally capable toy. The next step, then, was to add new capabilities that allow the user to better exploit the unique offerings of Python running natively on a GPU.

3.2.1. A GPU THREAD BARRIER

CUDA offers the programmer the `__syncthreads()` intrinsic: a hardware-enforced barrier where each CUDA thread will wait until all threads in the same thread block have arrived

and synced. We make this functionality directly available in *gpusnek* via the `syncthreads()`, which appears to be a globally-available function but is in fact a custom lexer and parser rule that emits a new dedicated instruction in the compiled bytecode. The example in Section 2.2 demonstrates using this functionality to enable threads to work together on shared data.

3.2.2. STDIN AND STDOUT

gpusnek offers multiple IO modes that can either cyclically buffer input/output in dedicated per-thread memory regions or use CUDA's included `printf` functionality to print directly from the kernel to the hosts terminal. This makes it more flexible and useful as a component to be incorporated into other projects.

3.2.3. AN IN-MEMORY VIRTUAL FILESYSTEM

We have included in *gpusnek* a port of MicroPython's virtual filesystem implementation, as well as a port of the specific filesystem *littlefs2* [2]. *littlefs2* is designed to run on micro-controllers and therefore boasts the following qualities that make it perfect for running on a low-resource GPU thread:

- Strictly bounded RAM usage as the contents of the filesystem grows.
- No dynamic memory allocations (can be provided with statically allocated buffers for working memory).
- No unbounded recursion, making maximum stack sizes computable.

With these implementations available on the GPU, we are able to format an in-VRAM buffer as a *littlefs2* disk image and mount it, all from within Python. If multiple threads mount the same buffer, they can access a shared filesystem from separate interpreters concurrently, even supporting simultaneous reads. Simultaneous writes are not supported, though we implement no mechanism to prevent them; race conditions in user code may silently corrupt not only the contents of the written file, but also the entire disk image :). Since the filesystem lives in global VRAM it can persist between kernel launches for the duration of the program. To persist it longer, we can also set up a mechanism to copy the disk image back to the host at shutdown, save it to the host disk, then reload it on subsequent launches. The host could even choose to mount the disk image itself, allowing for a filesystem shared between both the host CPU and the GPU (assuming an application-appropriate synchronization mechanism is put in place).

The examples in Sections 2.3 and 2.4 demonstrate the filesystem in action.

4. Performance

Not great, but honestly not as bad as you might expect!

5. Conclusion

In this paper we introduced *gpusnek*, a functional port of the MicroPython lexer, parser, compiler and interpreter to CUDA, enabling GPU-native execution of familiar Python code at GPU scales of parallelism. We implement not only those components that appear to be a good fit for both Python and CUDA but also those that are normally the domain of just one of the other, such as thread-synchronization primitives and filesystem accesses. With these tools, we bring all the familiar upsides and downsides of Python to the GPU, and even discover some new ones! In particular, we demonstrate that it is possible to interactively drive 1 Million Python interpreters at the command line on consumer hardware, and use them to parallelize non-trivial* tasks. The full source code for *gpusnek* is made available.

<https://github.com/jndean/gpusnek>

References

- [1] Micropython. www.micropython.org. Accessed: 31-03-2026.
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